

Experiment 5. Case study on Simulation of Engineering system

Aim: To design and simulate a model for speed control of a **DC motor using MATLAB Simulink**, and to analyze the effect of varying armature resistance on motor speed.

Procedure:

1. Open Simulink

Launch MATLAB → Open Simulink → Create a new blank model.

2. Add Required Blocks

DC Voltage Source

DC Motor block

Resistors (R1, R2, R3)

Switches (for varying resistance)

Step input blocks (to control switches)

Gain block and Scope (for output visualization)

3. Build the Circuit

Connect resistors in series with the armature circuit.

Attach switches across resistors to include/exclude them from the circuit.

Connect the DC motor terminals (A+, A-, F+, F-) properly.

Provide constant field excitation.

Connect mechanical load torque input (TL).

4. Control Mechanism

Use step signals to operate switches at different times.

This changes the effective armature resistance.

5. Measurement Setup

Connect motor speed output to a gain block and then to a scope.

6. Simulation Settings

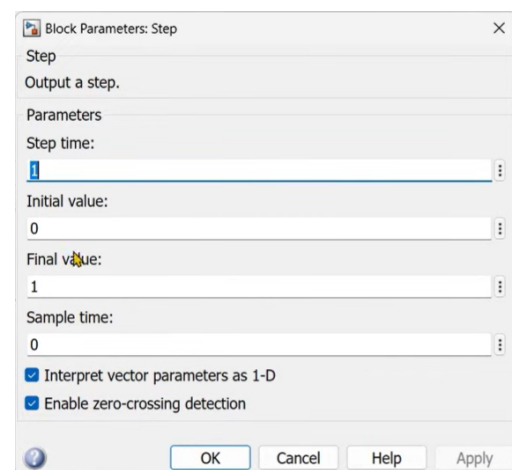
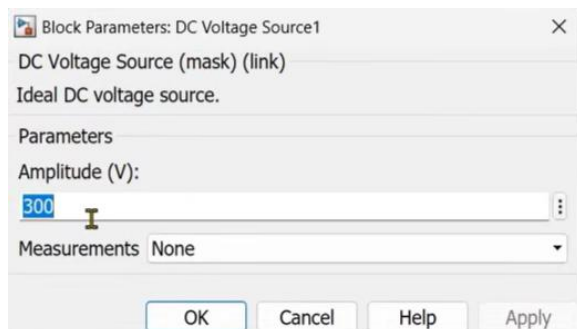
Set simulation time (e.g., 5–10 seconds).

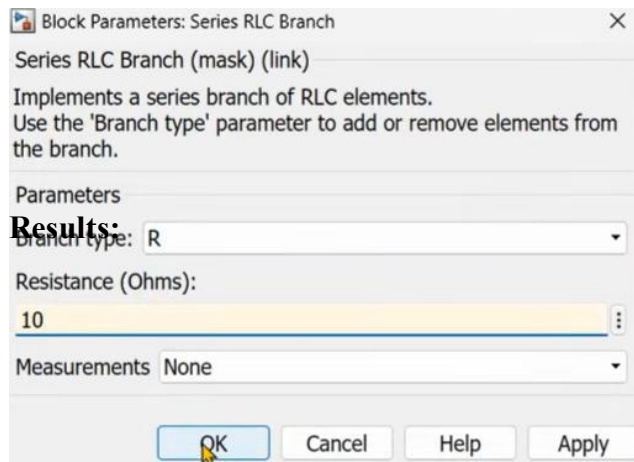
Choose appropriate solver (e.g., ode45 or fixed-step).

7. Run Simulation

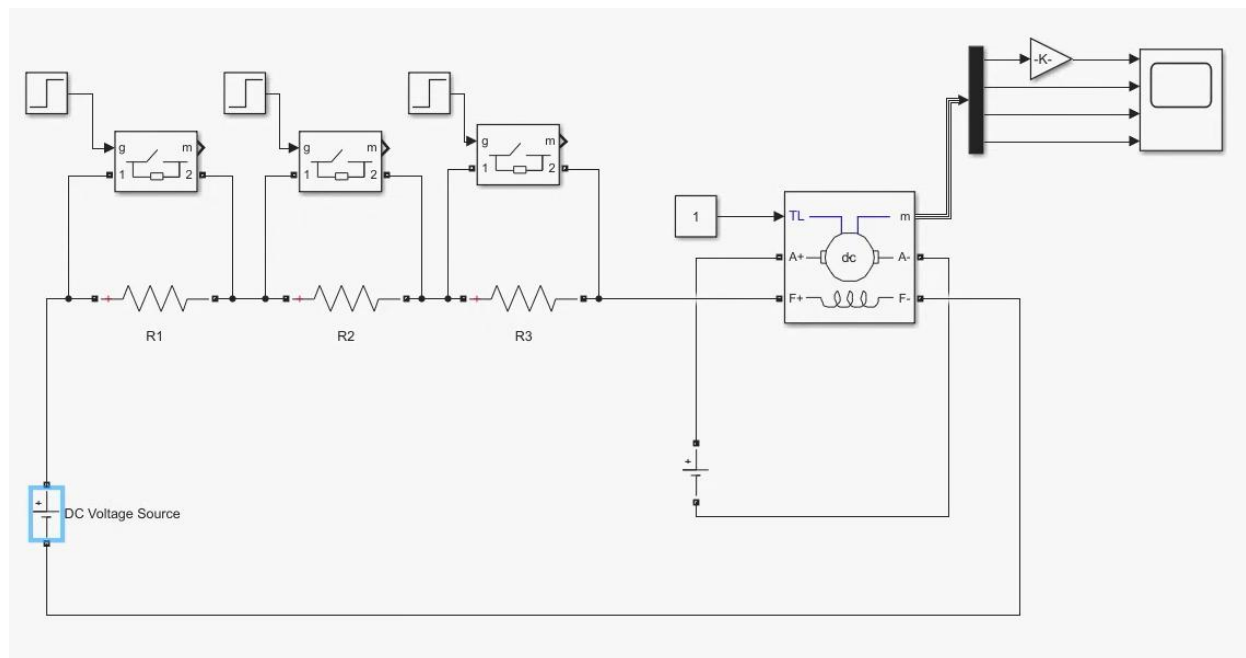
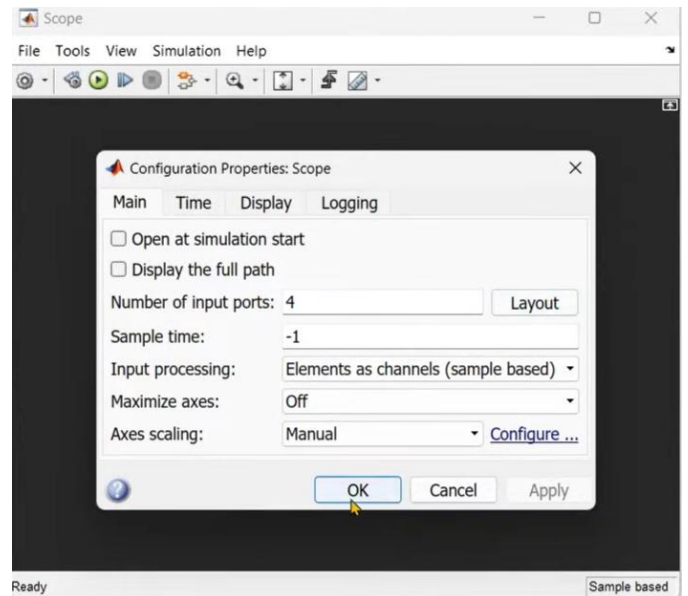
Observe speed variation on the scope as resistance changes.

Input.





Results:





Conclusion:

The simulation demonstrates that speed of a DC motor can be controlled by varying the armature resistance.